Hypothesis Testing for the Population Proportion of a Binomial Distribution

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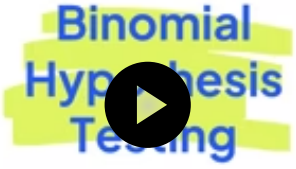

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Binomial hypothesis testing

How do I carry out a hypothesis test using a binomial distribution?

- The **population parameter** being tested will be the **probability, p** in a **binomial distribution** $B(n, p)$
- A hypothesis test is used when the assumed probability is questioned
- The **null hypothesis**, H_0 and **alternative hypothesis**, H_1 will always be given in terms of p
  - Make sure you clearly define p before writing the hypotheses
  - The null hypothesis will always be $H_0 : p = \ldots$
  - The alternative hypothesis will depend on if it is a one-tailed or two-tailed test
  - A one-tailed test would test to see if the **value of p** has **either increased or decreased**
    - The **alternative hypothesis, H_1 will be $H_1 : p > \ldots$ or $H_1 : p < \ldots$**
  - A two-tailed test would test to see if the **value of p** has **changed**
    - The **alternative hypothesis, H_1 will be $H_1 : p \neq \ldots$**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- When defining the test statistic, remember that the value of p is being tested, so this should be written as p in the original definition, followed by the null hypothesis stating the assumed value of p
- The binomial distribution will be used to **calculate the probability** of the test statistic taking the observed value **or a more extreme value**
- The hypothesis test can be carried out by
  - either calculating the probability of the test statistic taking the observed or a more extreme value (**p – value**) and comparing this with the significance level
  - or by finding the critical region and seeing whether the observed value of the test statistic lies within it
    - Finding the critical region can be more useful for considering more than one observed value or for further testing

How do I find the critical value?

- The **critical value** will be the **first value** to fall within the **critical region**
  - The binomial distribution is a **discrete** distribution so the probability of the observed value being within the critical region, given a true null hypothesis may be less than the **significance level**
  - This is the **actual significance level** and is the probability of **incorrectly** rejecting the null hypothesis

- For a **one-tailed test** use the tables of the binomial distribution or your calculator to find the first value for which the probability of that **or a more extreme value** is less than the given significance level
  - Check that the next value would cause this probability to be greater than the significance level
    - For $\mathbf{H_1}: p < \ldots$ if $\mathrm{P}(X \leq c) \leq \alpha\%$ and $\mathrm{P}(X \leq c+1) > \alpha\%$ then c is the critical value
    - For $\mathbf{H_1}: p > \ldots$ if $\mathrm{P}(X \geq c) \leq \alpha\%$ and $\mathrm{P}(X \geq c-1) > \alpha\%$ then c is the critical value

- For a **two-tailed test** you will need to find **both critical values,** one at each end of the distribution
  - Use the tables of the binomial distribution or your calculator to find the first value for which the probability of that **or a more extreme value** is less than **half** of the given significance level in both the upper and lower tails
  - Often one of the critical regions will be much bigger than the other

How do I lay out the hypothesis test?

- **Step 1.** Define the probability, p
- **Step 2.** Write the null and alternative hypotheses clearly using the form

$$H_0 : p = \ldots$$

$$H_1 : p = \ldots$$

- **Step 3.** Define the test statistic, usually $X \sim B(n,p)$ where n is a defined number of trials and p is the population

- **Step 5.** Compare the observed value of the test statistic with the critical value(s) or the p - value with the significance level
- **Step 6.** Decide whether there is enough evidence to reject H_0 or whether it has to be accepted
- **Step 7.** Write a conclusion in context

Worked Example

Jacques, a breadmaker, claims that more than 80% of people that shop in a particular supermarket buy his brand of bread. Jacques takes a random sample of 100 customers that have purchased bread and asks them which brand of bread they have purchased. He records that 86 of them had purchased his brand of bread. Test, at the 10% level of significance, whether Jacques' claim is justified.

Answer:

Step 1: Define the probability, p
Let p be the proportion of people who purchase Jacques' brand of bread.

Step 2: Write the hypotheses
$H_0: p = 0.8$
$H_1: p > 0.8$

Step 3: Define the test statistic
For $X \sim B(100, p)$

Step 4: Calculate the p-value
Assuming H_0 is true, then $P(X \geq 86) = 0.08044$

Step 5: Compare the p-value with the given significance level
$0.08044... < 10\%$

Step 6: Decide whether there is enough evidence to reject H_0
The p-value is less than the significance level so there is enough evidence to reject H_0

Step 7: Write a conclusion in context
There is sufficient evidence to reject the null hypothesis at the 10% level of significance.

The test supports Jacques' claim that more than 80% of people who shop in the supermarket buy his brand of bread.

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* Note: If finding the critical region, steps 4 and 5 would be:

Step 4: Calculate the critical value
Assuming H_0 is true, then $P(X \geq 86) = 0.08044... < 10\%$ and $P(X \geq 85) = 0.1285... > 10\%$ So the critical value is 86 and the critical region is $X \geq 86$

Step 5: Compare the observed value of the test statistic with the critical region:
86 is inside the critical region.

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Examiner Tips and Tricks

- Make sure you are careful with the inequalities when finding critical values from the binomial tables or from

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